Geometry & Measurement

ANSWER KEY

Section 1: Angles & Triangles

1 In a triangle, the angles are in the ratio $2:3:5$. What is the measure of the largest angle?

a. 90° ✓

b. 100°

c. 80°

d. 75°

Total parts $= 2+3+5 = 10$. The sum of angles in a triangle is 180°. One part $= \frac{180}{10} = 18^\circ$. The largest angle $= 5 \times 18 = 90^\circ$.

2 In a right triangle, one leg is 12 cm and the hypotenuse is 13 cm. What is the area of the triangle?

a. $30\ \text{cm}^2$ ✓

b. $60\ \text{cm}^2$

c. $39\ \text{cm}^2$

d. $25\ \text{cm}^2$

By Pythagorean triple $5-12-13$, the second leg is $\sqrt{13^2-12^2} = 5$ cm. Area $= \frac{1}{2} \times 12 \times 5 = 30\ \text{cm}^2$.

3 Two parallel lines are intersected by a transversal. If an interior angle measures $(3x-15)^\circ$ and its consecutive interior angle measures $(2x+20)^\circ$, what is x?

a. 35 ✓

b. 30

c. 40

d. 25

Consecutive interior angles are supplementary:
$(3x-15)+(2x+20) = 180 \Rightarrow 5x+5 = 180 \Rightarrow 5x = 175 \Rightarrow x = 35$.

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Section 2: Circles, Circumference & Area

4 A circle has a circumference of 16π cm. What is its area?

a. 64π cm^2 ✓

b. 32π cm^2

c. 16π cm^2

d. 128π cm^2

Circumference $= 2\pi r = 16\pi \Rightarrow r = 8$ cm. Area $= \pi r^2 = \pi(8^2) = 64\pi$ cm^2.

5 A square of side length 10 cm has a circle inscribed within it. What is the area of the shaded region between the square and the circle?

a. $100 - 25\pi$ ✓

b. $100 - 10\pi$

c. $50 - 25\pi$

d. $100 - 50\pi$

Square area $= 10^2 = 100$. The inscribed circle diameter is 10, so $r = 5$, and circle area $= \pi(5^2) = 25\pi$. Shaded area $= 100 - 25\pi$.

6 If the radius of a circle is doubled, by what factor does its area increase?

a. 4 ✓

b. 2

c. 8

d. 16

Area is proportional to r^2. Doubling the radius multiplies the area by $2^2 = 4$.

Section 3: Quadrilaterals & Polygons

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7 A rectangle has a perimeter of 40 cm and its length is 3 times its width. What is the area of the rectangle?

a. $75\ \text{cm}^2$ ✓

b. $80\ \text{cm}^2$

c. $100\ \text{cm}^2$

d. $60\ \text{cm}^2$

Perimeter $= 2(l + w) = 40 \Rightarrow l + w = 20$. Since $l = 3w$, we have $4w = 20 \Rightarrow w = 5$, and $l = 15$. Area $= 15 \times 5 = 75\ \text{cm}^2$.

8 What is the sum of the interior angles of a regular hexagon?

a. 720° ✓

b. 540°

c. 360°

d. 900°

Formula for sum of interior angles: $(n - 2) \times 180^\circ$. For a hexagon ($n = 6$): $(6 - 2) \times 180 = 4 \times 180 = 720^\circ$.

9 A rhombus has diagonals of lengths 12 cm and 16 cm. What is the length of one side of the rhombus?

a. 10 cm ✓

b. 8 cm

c. 14 cm

d. 20 cm

The diagonals bisect each other at right angles into half-lengths of 6 cm and 8 cm. By the Pythagorean theorem: Side $= \sqrt{6^2 + 8^2} = \sqrt{100} = 10$ cm.

Section 4: Solid Geometry & Volume

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10 A rectangular tank has dimensions $4\text{ m} \times 5\text{ m} \times 6\text{ m}$. If it is filled with water at a rate of $10\text{ m}^3/\text{min}$, how many minutes does it take to fill completely?

a. 12 **minutes** ✓

b. 10 minutes

c. 15 minutes

d. 20 minutes

Volume $= 4 \times 5 \times 6 = 120\text{ m}^3$. Time $= \frac{120}{10} = 12$ minutes.

11 A cylinder has base radius 3 cm and height 10 cm. What is its volume in cubic centimeters?

a. 90π ✓

b. 60π

c. 30π

d. 180π

Cylinder volume $= \pi r^2 h = \pi(3^2)(10) = 90\pi\text{ cm}^3$.

12 If the volume of a cube is 64 cm^3, what is its total surface area?

a. 96 cm^2 ✓

b. 64 cm^2

c. 128 cm^2

d. 48 cm^2

Side length $s = \sqrt[3]{64} = 4\text{ cm}$. Total surface area $= 6s^2 = 6(4^2) = 6 \times 16 = 96\text{ cm}^2$.

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